

Ashmit Dutta

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science, Computer Engineering; Minors in Math & Physics

Aug 2023 – May 2027

- Accepted paper at ICML 2025 workshop on [finetuning Chain of Thought in LLMs](#)

EXPERIENCE

Software Engineer, Part-Time Student

Jan 2025 – Jan 2026

John Deere

Champaign, IL

- Shipped a React + Spring Boot data-exporting platform **to production** for **500+ internal users**.
- Built core REST APIs and CI/CD pipelines over PostgreSQL and DB2 handling **millions of records**.
- Added AWS CloudWatch observability cutting issue triage time, and secured sensitive export data in S3.

Machine Learning Research Intern (Prof. Huan Zhang)

Dec 2025 – Present

Assured and Trustworthy AI Research Lab - University of Illinois

Urbana, IL

- Shipping Python and CUDA features to [auto.LiRPA](#), the core verification library behind α, β -CROWN (**VNN-COMP 2021–2025 winner**) used for proving safety of learned controllers and perception models.
- Building evaluation pipelines and benchmark datasets for the lab's VNN-COMP 2026 global submission.

Machine Learning Research Intern (Fermilab collaboration)

Jun 2024 – Aug 2024

University of Chicago Data Science Institute — DSI Summer Lab

Chicago, IL

- Selected for DSI's competitive summer research program; built a reusable multi-modal Graph Neural Network library in PyTorch Geometric for Fermilab's Exa.TrkX neutrino-reconstruction project.
- Engineered an end-to-end training pipeline over **20,000+ detector events** (H5Py) on a Slurm HPC cluster; presented results at the DSI summer research symposium.

PROJECTS

[Stanford TreeHacks: Multi-Modal Video Search Engine](#) | *Python (Modal, BrightData, FastAPI)*

Feb 2026

- Winner (Modal Inference Track + Bright Data AI) at Stanford TreeHacks 2026** (out of 380 teams).
- Built a distributed full-stack video search and dataset-generation app on cloud GPU inference.
- Designed a multi-modal embedding pipeline fusing vision and audio for semantic retrieval, indexing 24 hr+ videos.
- Integrated Bright Data scraping agents for automated data ingestion; deployed on Modal with MongoDB.

[GPU Accelerated Deep Learning Inference \(LeNet-5\)](#) | *C++, CUDA*

Feb 2026 – Apr 2026

- Wrote shape-specialized implicit-GEMM CUDA kernels with fused im2col and constant-memory weights.
- Accelerated Conv2 with WMMA Tensor Cores for **6.6× speedup**; 12.5 ms inference at batch 10,000.
- Drove design via Nsight Compute roofline/stall analysis; **placed 5/130 in ECE 408 competition**.

[Autonomous Drone Racing](#) | *ROS, Python, C++, PyTorch*

Oct 2025 – Dec 2025

- Built a ROS 2 stack for a Crazyflie 2.1 drone using a min-snap planner and geometric controller.
- Tuned controller gains via Bayesian optimizer; **outperformed 3 published baselines** on real hardware.
- Trained a YOLOv8 gate-segmentation model on NeRF-rendered data (**98% accuracy**).
- Analyzed flight Rosbags to debug tracking error; designed a voltage-battery feedback system to prevent thrust sag.

LEADERSHIP, PUBLICATIONS & AWARDS

Co-Founder & Lead Organizer, [Online Physics Olympiad \(OPhO\)](#): largest student-run international physics competition — built a **22k+ Discord community**, 3k+ Facebook following, and **3k peak annual competitors** since 2020; raised **\$10k+ in sponsorships** from Citadel, Jane Street, and Wolfram.

Awards & Leadership: 2nd Place, Bradley Mottier Innovation Challenge, UIUC ISE (\$2,000) — AR glasses for ASL translation. Executive Board Member (Treasurer), IEEE Chapter at UIUC.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, SystemVerilog, RISC-V Assembly

ML & AI: PyTorch/Geometric, CUDA, Modal, LLM APIs, embeddings, multi-modal retrieval

Backend & Infra: REST APIs, FastAPI, Spring Boot, PostgreSQL, DB2, MongoDB, Docker, Git, CI/CD, AWS, Linux

Systems & Hardware: RTL design, FPGAs, STM32/PCB design, embedded firmware, GDB, QEMU